

A 1596-GB/s 48-Gb Stacked Embedded DRAM 384-Core SoC With Hybrid Bonding Integration

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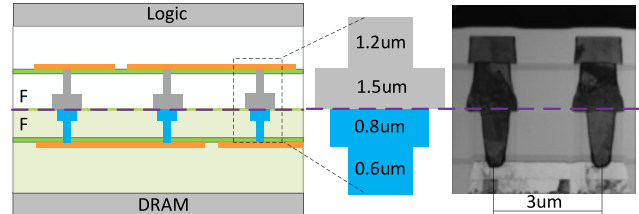
Abstract—A 384-Keccak cryptographic cores system on chip (SoC) (die size $> 800 \text{ mm}^2$ and reticle size $> 25.5 \text{ mm} \times 32.0 \text{ mm}$) with 48-Gb stacked embedded DRAM (SeDRAM) is presented using 110-K/mm^2 integration density and 0.1-fF capacitive load hybrid bonding integration. The SeDRAM's advantages of low power and low latency are fully utilized by 49 152 wide-IO and network-on-chip design in logic. Thermal stress and reliability are resolved to get a high package yield. The proposed SoC consumes 1596-GB/s bandwidth with 55-W power consumption only and achieves a great system performance improvement.

Index Terms—Bandwidth, hybrid bonding (HB), performance, stacked embedded DRAM (SeDRAM), system on chip (SoC).

I. INTRODUCTION

High-bandwidth and high energy efficiency DRAM is one of the most important limiting factor for emerging high-performance system-on-chip (SoC) applications. To meet the demand of high-bandwidth, GDDR and high-bandwidth memory (HBM) provide alternative solutions. An 8-nm 18-Gb/s GDDR6 PHY is presented in [1], which means the single chip bandwidth is 72 GB/s. To get higher bandwidth, multiple GDDR DRAMs can be used, which will involve an issue of high-speed signal routing on PCBs. HBM uses multiple DRAM chip stacks [2] and 1024 I/Os facilitated by a 2.5-D silicon interposer and through silicon via (TSV) to obtain 640-GB/s bandwidth [3]. However, HBM's major obstacle for achieving high bandwidth is the heavy capacitive load due to TSVs and silicon interposer [4]. Without the interposer, the 3-D stack offers high density and low-power signaling between dies. The computing performance improvement is ensured by an 8.8- μm pitch hybrid bonding (HB) 3-D integration with memory [5]. Our previous work [6] reported 136-GB/s 4-Gb LPDDR4/4X logic-to-memory interface with HB integration.

Fig. 1 shows the used HB integration and its performance benchmark. As an extension of the back-end-of-line (BEOL) process, HB connects two dies with Cu–Cu metal bonding at temperature below 350°C . The bonding flow is previously mentioned in [6]. Wafer-level face-to-face (F2F) integration enables a short connection wire



Typical 3D scheme performance benchmark

Technology	3D-IC F2F	SoIC F2F	HB F2F
Bump density	1.0x	16.0x	160.0x
Speed (1/wire delay)	1.0x	11.9x	59.5x
Bandwidth density (bump density $\times$ speed)	1.0x	19.1x	955.0x
Power efficiency (energy/bit)	1.0x	0.05x	0.02x

Fig. 1. Used HB and performance benchmark.

$\sim 2 \text{ }\mu\text{m}$ with a low latency and a 0.1-fF low capacitive load. The benchmark comparison in gray colored from [7] shows the HB integration in this work offers $50\times$ better bandwidth density and $2.5\times$ better power efficiency. The wafer accept test (WAT) result shows the HB interconnection pass ratio is above 99.99999%. Moreover, the twin HB structure is employed to improve the interconnection reliability.

In this work, 48-Gb stacked embedded DRAM (SeDRAM) die and 384-core logic die are designed and integrated with HB to achieve a better computing performance. In the following, Section II introduces the proposal of SoC. Section III presents the design of SeDRAM structure and interface timing. Section IV describes the architecture of the logic circuit. Section V shows the integration of two wafers. The silicon measurement results are provided in Section VI. Finally, this work is summarized in Section VII.

II. OVERVIEW OF PROPOSED SOC CHIP

As shown in Fig. 2, the proposed SoC chip is flip-chip structure. Die size $> 800 \text{ mm}^2$ is accommodated by a package size 2500 mm^2 with making full use of the scribe line. As internal signals, SeDRAM's data and command are connected to logic by 3- μm pitch wafer-on-wafer (WoW) HB with 110-K/mm^2 integration density, which is benefit to wide-IO data interface.

To address the issue of thermal stress of such large chip, dummy bumps are used in the substrate design. The real bump is embedded into Si of logic die to PAD out from backside. The smaller size dummy bump location on Pi is to balance the height offset with real bump on the reflow process. To avoid hot spot, the distribution of ball with max current is uniform, which will improve the reliability of substrate. Furthermore, the alternating current (ac) impedance of power distribution network (PDN) is reduced for a low noise. The VSS balls are assigned under the chip for excellent thermal dissipation.

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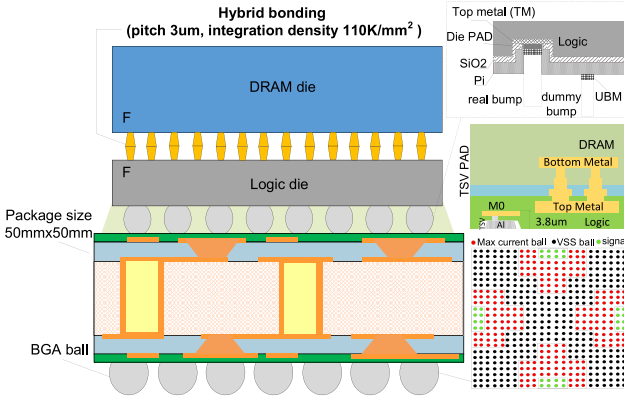


Fig. 2. Proposed flip-chip SoC structure.

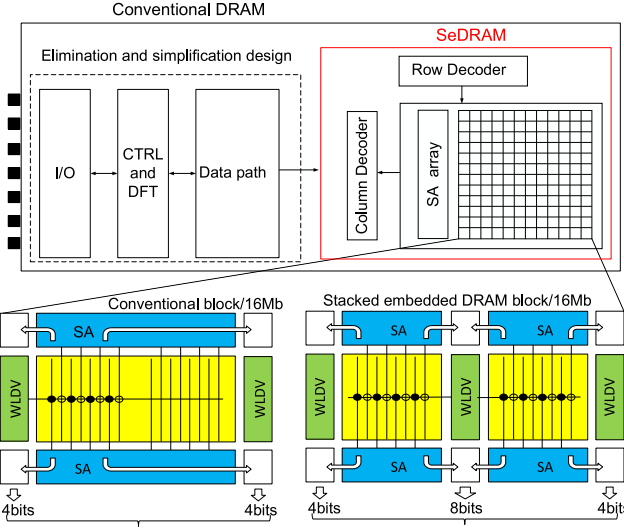


Fig. 3. Difference of conventional DRAM and SeDRAM.

III. STACKED EMBEDDED DRAM DESIGN

Fig. 3 shows the architecture difference between SeDRAM and conventional DRAM. Compared to the conventional DRAM, the circuit of I/O analog, CTRL, and data path are eliminated and the design for test (DFT) circuit is migrated to logic. These optimization designs greatly save power consumption and reduce the latency of access array because of removing the casing high-power and high-latency circuit on the data channel. In addition, the parallelism of access SeDRAM is improved by the extended wide-IO data interface from logic to the array.

Different with the conventional DRAM block, the SeDRAM extends data-width to 16 bits by a shared word line driver (WLDV). It enables 128-bits data-width interface for SeDRAM memory bank (128 Mb). With the 110-K/mm² HB interconnection, the data width of 48-Gb SeDRAM is 49152 ($128 \times 8 \times 48$) bits between the computing logic and array.

Fig. 4 shows the access interface timing of SeDRAM. Since the interface runs at the clock rate of several hundred megahertz with shortened data line, it has advantages of low power consumption and low latency. After the page is activated by the bank selection signal and row address accordingly, write or read operation is triggered by write and read clocks, respectively. The write or read clock is 3.76 ns in period, which means the data rate per pin is 266 Mbps. Therefore,

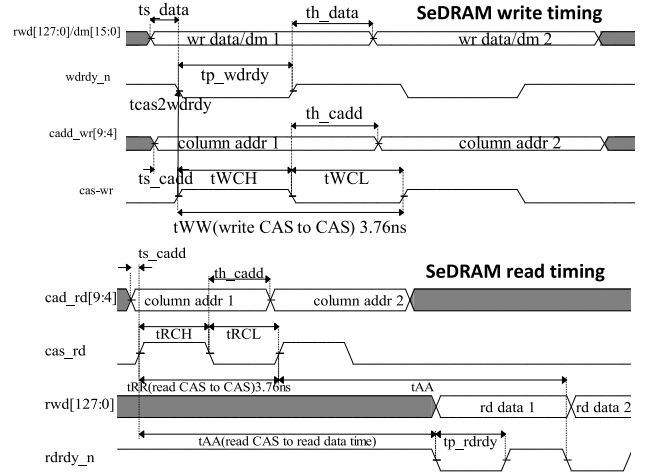


Fig. 4. SeDRAM interface timing.

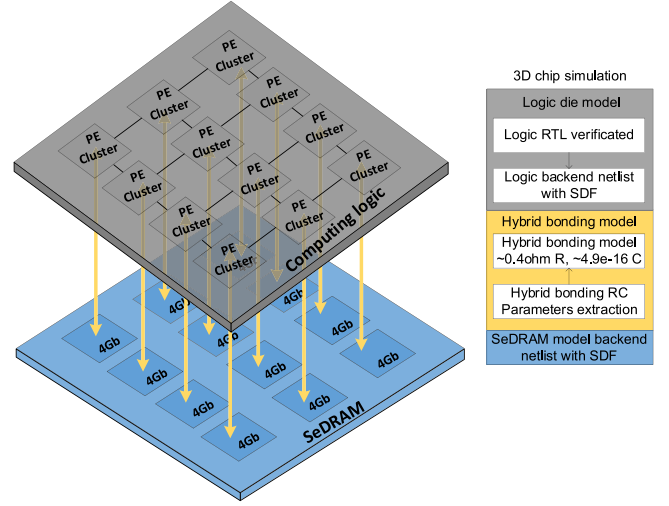


Fig. 5. Diagram of computing logic and SeDRAM.

the chip memory bandwidth is 1596 GB/s ($266 \text{ Mbps} \times 49152$) with 48-Gbits SeDRAM.

IV. COMPUTING LOGIC ARCHITECTURE

Fig. 5 shows the computing logic is a 4×3 mesh network on chip (NoC) with 12 PE clusters in the SoC. Each PE cluster consists of 32-Keccak cryptographic core.

The 48-Gb SeDRAM has 12 partitions corresponding to 12 PE clusters, and each partition has 4-Gb capacity. A cycle-accurate 3-D chip-level simulation, including the logic die model, HB model, and SeDRAM model, is built to verify the chip function and timing. The HB model uses the real RC parameters for precision latency on the data path of access SeDRAM.

Fig. 6 shows the two data paths of computing logic access SeDRAM. As local memory, logic computing core accesses its self SeDRAM partition using a router transparent transmission command to the memory controller (MC) via a buffer and HB connection to the array. The core can also access other SeDRAM partitions as global memory. In this case, the command is transfer to the destination router and then access is executed by local memory, and finally, the operation result is transited to the source.

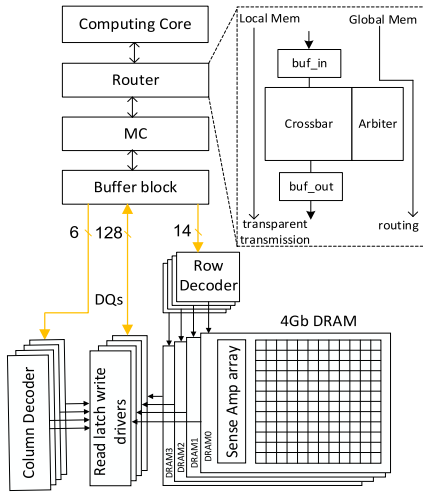


Fig. 6. Computing core access local and global memory.

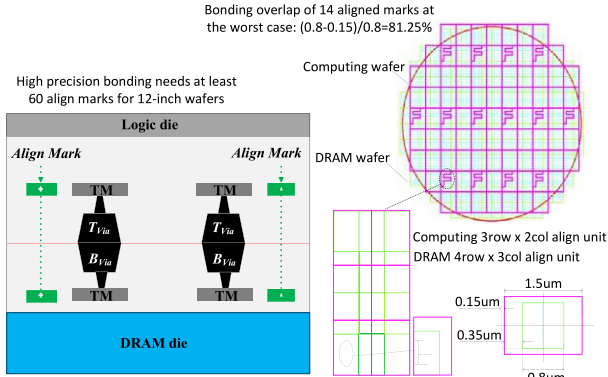


Fig. 7. Align mark and bonding precision.

V. TWO DIE INTEGRATION

Fig. 7 shows the HB integration alignment between the computing logic die and SeDRAM die. Align mark is used for an accurate HB interconnection of logic and SeDRAM wafers. The position of align marks need to be computed and printed onto the computing logic and SeDRAM graphic data system (GDS) in advance. A precision alignment needs at least 60 marks for 12-inch wafer, which result in area cost of GDS. To reduce the area cost of GDS, the 14 aligned marks on wafers provide 81.25% bonding overlay at the worst case, whose electrical performance and reliability have been proved by silicon.

The thermal stress is considered as a key package issue for such larger chip, which has an important impact on yield. To reduce the thermal stress, the ~ 86 million dummy HBs ($\sim 98\%$ of HBs in this chip) are employed to get a more uniform stress distribution between logic wafer and SeDRAM wafer. The simulation results show that the maximum thermal stress is reduced 15% by using dummy HB than that without, as shown in Fig. 8.

The SoC chip power consumption is 55 W when 384-core computing consumes 1596-GB/s bandwidth. The chip thermal distribution is almost the same for single and stacked dies with the dominant conditions of 10-L PCB, enhanced heatsink, and forced air 40 CFM, which indicates the employed 3-D structure has a limited influence on junction temperature for the specific application. The maximum junction temperature 77.70°C meets the requirement of DRAM process, as shown in Fig. 9.

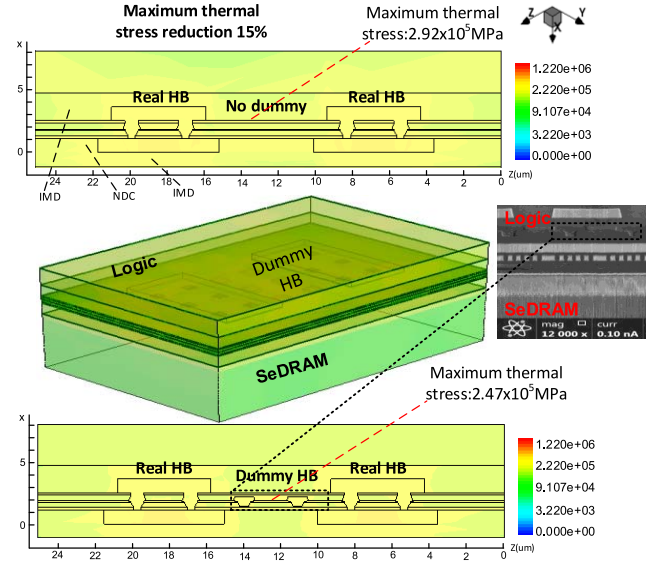


Fig. 8. Stress results with dummy HB.

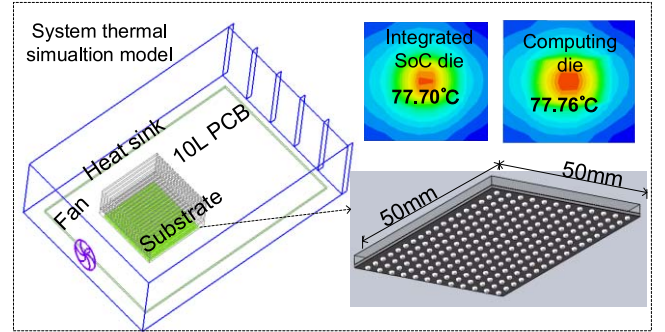


Fig. 9. System thermal simulation.

TABLE I
RELIABILITY TEST CONDITIONS AND RESULTS

Item	Condition	Result
T/C	$-65^\circ\text{C}\sim 150^\circ\text{C}$, 1000cycles, 2cycles/hr	PASS
THS	$85^\circ\text{C}/85\%\text{RH}$, 1000cycles	PASS
HTS	150°C , 1000hr	PASS
EM	$T=250\sim 350^\circ\text{C}$, $J=1\sim 5\text{MA}/\text{cm}^2$	PASS
IMD_Vramp	Linear voltage ramp, $V_{\text{max}}=200\text{V}$, $V_{\text{step}}=0.1\text{MV}/\text{cm}$	PASS

VI. MEASUREMENT RESULTS

Fig. 10 shows the measured results. The shmoo plot shows that the access speed is 266 Mbps/pin, which means the SoC's memory bandwidth is up to 1596 GB/s. This bandwidth is about two times HBM2E. The implemented energy efficiency 0.88 pJ/b is about 6.67 times better than HBM2E.

JEDEC qualification tests are performed to ensure the process meets industry reliability standards. The reliability test results of temperature cycling (T/C), temperature humidity stress (THS), high temperature stress (HTS), electromigration (EM), and additional IMD_Vramp in Table I show all qualification items passed with the margin.

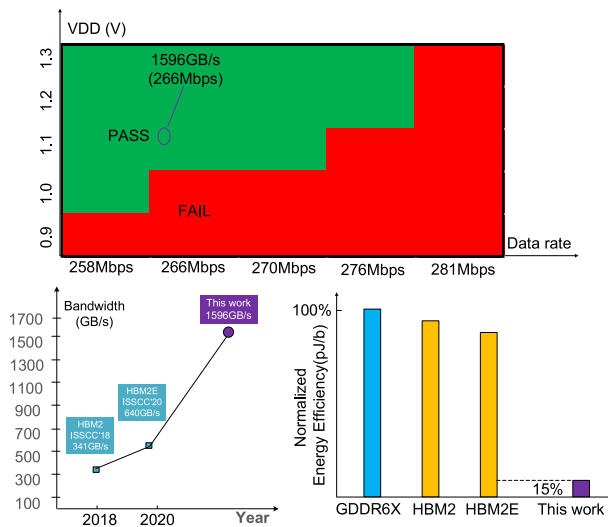
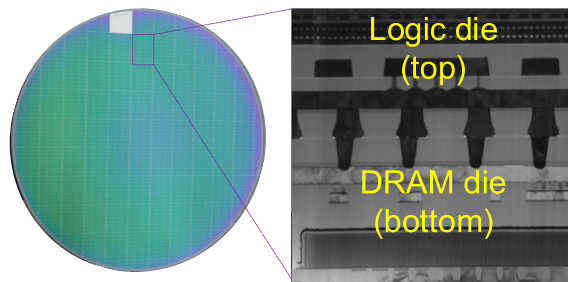


Fig. 10. Chip bandwidth and energy efficiency.



Chip Keccak performance comparison

Parameters	17JSSC[9]	This work
Technology node	130nm	25nm for DRAM, 55nm for logic
Clock	433MHz	400MHz
Cycles	1624	312
Throughput(kb/s)	827.59	1394871.79
Area(kGE)	7.5	39.1
Power (uW/MHz)	41.2	26.4
Gate efficiency (kbps/kGE)	109.8	35673.9

Fig. 11. Chip wafer, micrograph, and summary.

The chip wafer, performance comparison, and the cross-sectional TEM image are shown in Fig. 11. Many bandwidth-hungry applications, such as the secure hash algorithm (SHA) for data encryption, require short read access time (tAC) and wide-IO from memory [8]. The Keccak application achieves 1686 times throughput and two times power efficiency than the reported work [9]. Owing

to the bandwidth of SeDRAM is fully utilized by computing logic core to achieve this better performance.

VII. CONCLUSION

We have successfully integrated a 48-Gb SeDRAM die a 384-core logic die using WoW HB technology. This wafer-level F2F integration uses 3-um pitch Cu-Cu HB with the advantages of high integration density and low capacitive load. The effective simplified design of SeDRAM offered up to 1596-GB/s bandwidth and 0.88-pJ/b energy efficiency. The challenges of such a large chip associated with the thermal stress and reliability were resolved by the package design for high yield. The implemented high bandwidth is consumed by 49 152 wide-IO and NoC in the computing logic die. The measurement results of chip showed that several times performance improvement was achieved with only 55-W power consumption.

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